



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 9 • STANDARD 6.NOS.7

Reflect Points Across Axes

Lesson 9-7 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can reflect points across the x-axis and y-axis on the coordinate plane.

Language Objective: I can explain my work using the words reflection, x-axis, y-axis, and symmetry.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Reflection

x-axis

y-axis

Symmetry

Integer



Tap any word to see what it means and a picture.

1 A flip of a point or shape over an axis to make a mirror image is a

2 The horizontal number line on the coordinate plane is the .

3 The vertical number line on the coordinate plane is the .

4 When two halves are mirror images of each other, the figure has

5 A whole number or its opposite, with no fraction part, is an .

Reflect — Exit Ticket

What is the reflection of $(-5, 2)$ over the y-axis?

- A. $(-5, -2)$
- B. $(5, -2)$
- C. $(5, 2)$
- D. $(2, -5)$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Reflection
2. **Notes 2:** x-axis
3. **Notes 3:** y-axis
4. **Notes 4:** Symmetry
5. **Notes 5:** Integer
6. **Watch for this mistake:** *A common mistake in Reflect Points Across Axes is skipping the key idea: "Reflect over the x-axis: the y-coordinate changes sign. Reflect over the y-axis: the x-coordinate changes sign." — always check your work against this rule before you submit.*
7. **Try It 1:** A. (5, 3) — *Reflecting across the y-axis flips the sign of the x-coordinate only. (-5, 3) becomes (5, 3).*
8. **Try It 2:** A. The y-coordinate — *Across the x-axis the rule is $(x, y) \rightarrow (x, -y)$, so only the y-coordinate flips sign. (8, -6) becomes (8, 6).*
9. **Exit Ticket:** C. (5, 2) — *Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -5 to 5, and y stays 2. The reflection is (5, 2).*